Adaptive Optics Lab

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December 14, 2024

1 Introduction

Adaptive optics is a technique for correcting optical distortions to improve image quality. In observational astronomy, these distortions come from the light from stars and galaxies passing through the atmosphere, and they blur images of stars and galaxies into big circles, so it is necessary to use adaptive optics to see finer details. However, the atmosphere is constantly changing, so it is also necessary to use a wavefront sensor to capture the light as it comes in and compare it to unobstructed light, creating a map of what corrections need to be made using an adjustable mirror in order to remove the distortions. In this lab, our goal was to create our own version of a wavefront sensor, specifically a Shack-Hartmann wavefront sensor, and use it to determine what kind of distortion is being caused by an unknown lens. A Shack-Hartmann wavefront sensor works by sending a collimated beam of light from a telescope's primary mirror through an array of small lenses (a lenlet array) and into an image detector, resulting in an image of a grid of focused spots. These spots slightly shift around based on the distortions of the incoming light, and they can be compared to a grid of spots from undistorted light so that an adjustable mirror can be manipulated accordingly until the spots from the incoming light match the positions of the spots from undistorted light. When the spots match up, the vast majority of the distortion will have been removed from the light reaching the telescope sensor. Below is a diagram of the image correction process of the Shack-Hartmann wavefront sensor (see Figure 1).

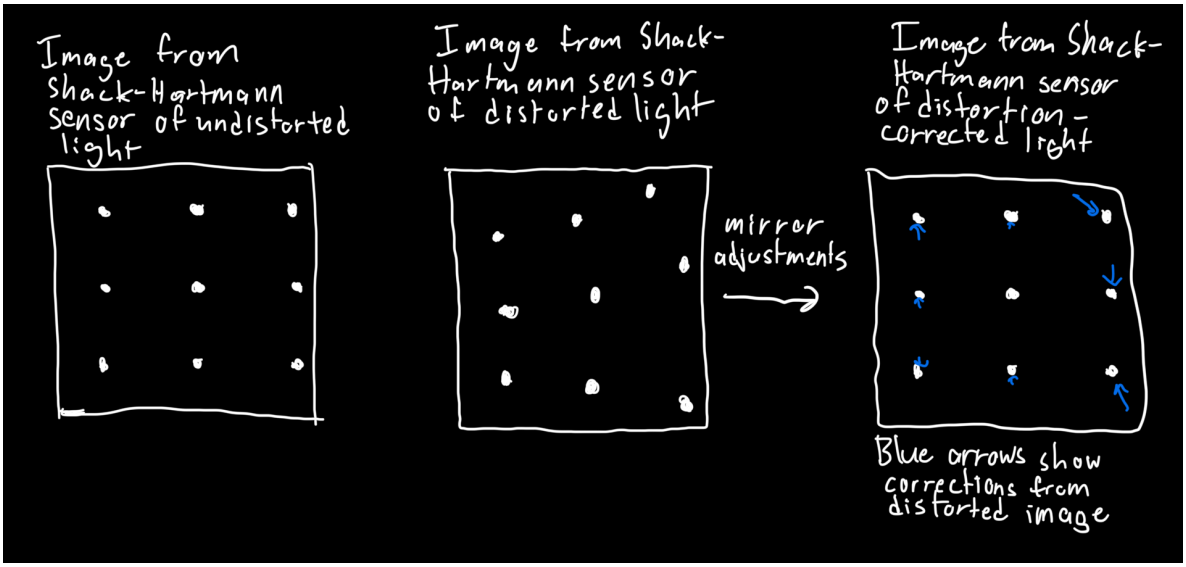


Figure 1: Image Correction Process of the Shack-Hartmann Wavefront Sensor

2 Experimental Setup

2.1 Simulating Starlight

Our first step in making the Shack-Hartmann wavefront sensor was to simulate starlight. To do this, we had to make a beam of light that appears like it is coming from infinitely far away, meaning the beam comes to a focus infinitely far away (collimated). We set up a lens with a focal length of 100 mm in front of the light source and moved it closer and farther from the source until we could verify that the beam of light past the lens was the same width at any distance. We verified this by marking the diameter of the beam on a card placed directly in front of the lens and then moving the card far down the beam and moving the lens slightly until the beam width matched the width just in front of the lens. This technique resulted in us placing the lens 106 mm in front of the source to get a collimated beam.

2.2 Simulating a Telescope

Our next step was to simulate a telescope looking at the starlight, which meant placing an adjustable aperture in front of the now collimated beam of light and another lens immediately after that to simulate a telescopes primary mirror and then placing an image detector at the focal point produced by this to represent the telescope's sensor. We found the focal point with the detector by moving it back and forth in the beam of light after the second lens until the spot of light in the image it produced was as small as possible. The distance we placed the detector at from the second lens ended up being 97 mm, which meant that this was where the focal plane was. We then adjusted the aperture to see how it affected the image on the detector, and we found that a smaller aperture let in less light, requiring a longer exposure time and producing a lower resolution image, while a larger aperture let in much more light, requiring less exposure time and increasing resolution, but it also introduced bloom and other distortions on the image. On a real telescope, this corresponds to the size of the primary mirror, as this controls how much light is reflected into the telescope.

2.3 Setting up the Location for the Lenslet Array

Our next goal was to find a location for the lenslet array, which had to be placed in a collimated beam at an image of the aperture to make sure that the grid of spots produced by undistorted light is not distorted itself and so that the spots are in sharp focus and visible. In order to find where the image of the aperture was after placing a third lens after the focal place to collimate the beam again like we did in the first part, we used this equation (see Equation 1).

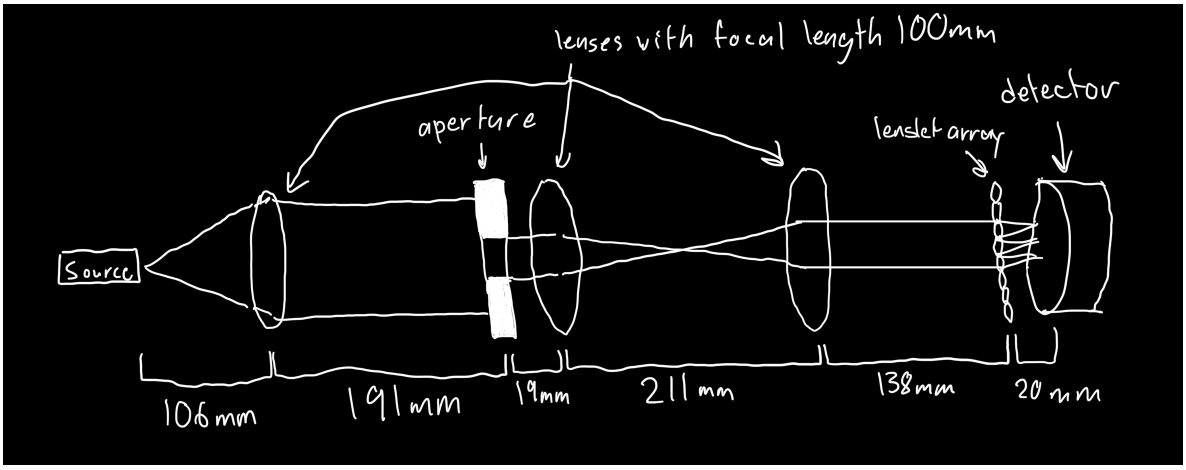


Figure 2: Final Experimental Setup

$$1/f = (1/image) - (1/object) \tag{1}$$

Where "f" is the focal length of the lens, "object" is the distance between the aperture and the third lens, and "image" is the distance from the third lens where the lenslet array should be placed. After placing the lenslet array at this position, we moved the detector around behind it and saw that the spots would lose focus as it got further away and came into a very sharp focus right in front of the lenlet array.

2.4 Measuring the Spot Outputs for the WFS

Specifically, the spots in the image produced by the detector came into the sharpest focus at a distance of twenty millimeters from the lenlet array, which led us to leave it there for our science images. We also found that we could see more spots on the detector if we opened the aperture more, but if you could not do this, like on a telescope, you could get a denser lenslet array with smaller lenses. Below is a diagram of the final experimental setup (see Figure 2).

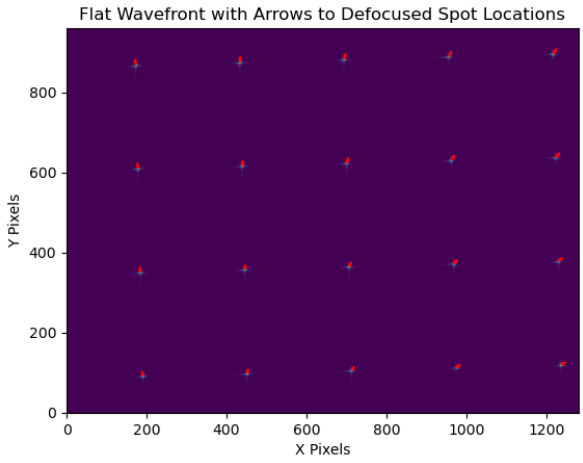


Figure 3: Flat Wavefront with Arrows to Defocused Spot Locations

3 Experimental Results

With this experimental setup, we captured images of the grid of spots, both undistorted and with an unkown lens between the first lens and the aperture. I then estimated the pixel locations of the centers of each spot by cropping the image down to each spot and estimating roughly where the middle is. I then used these locations as initial guesses in 2D Gaussian fits on each spot, giving me the exact locations of the spot centers for both the distorted and undistorted images. I then plotted arrows pointing from the undistorted spots to their distorted counterparts' locations on top of the undistorted image, resulting in the following image (see Figure 3). This image shows that the spots were mostly deflected radially outwards, meaning the unknown lens is a negative lens.

4 Conclusion

This lab helped me to understand the field of adaptive optics a little more, demonstrating how distortions in incoming light can warp an image and how to correct for this using a Shack-Hartmann wavefront sensor. My final image could be used to manipulate an adjustable mirror before the detector, correcting for the distortions. I could then remove the lenslet array and I would have an extremely clear image of the source even with the negative lens still there. On a real telescope, this same technique could be used to remove distortions from the atmosphere continuously, creating an image as if the atmosphere was not there, just like we could create an image as if the negative lens was not there.